

Adaptive Pairs Trading with Kalman Filter Signal Generation in HK Equities

A QUANTITATIVE ANALYSIS (25-26)

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PROJECT OVERVIEW - PURPOSE

1) Practicality of Statistical Arbitrage

Assess practical effectiveness of statistical arbitrage by testing whether strong in-sample performance remains robust when the strategy is applied to new, volatile, and less favourable market conditions.

- Evaluate performance when exposed to **higher volatility, changing relationships, and out-of-sample data**.
- **Investigate** whether historically identified pricing relationships **remain stable** when **market conditions shift**.

2) Replicability in Volatile Environment

This project aims to evaluate whether cointegration-based pairs trading remains effective in a relatively volatile equity market, using the Hong Kong stock market as a case study.

- We **assess** whether **cointegrated relationships** can generate robust **out-of-sample** trading performance.
- Use a **Kalman filter** to estimate dynamic hedge ratios that adapt over time.
- Evaluate whether combining **multiple statistical signals** improves the **viability** of a pairs trading strategy.

3) Refinement of Econometrics Knowledge

This project provides me with the opportunity to further deepen my knowledge in Econometrics and explore my passion for Trading and Global Markets.

- **Develop** econometric techniques introduced in **Financial Econometrics** module.
- **Deepen understanding** of how econometric methods can be **applied to real financial market data**.
- **Develop** more applied understanding of how **statistical models behave** when used in real-world market conditions.

PROJECT OVERVIEW - TIMELINE

May Data Collection

Collecting daily prices of equities from Bloomberg Terminal:

- Hang-Seng **Tech** Index
- Hang-Seng **Composite Energy**
- Hang-Seng **Composite Materials**

Filtered by trading volume > 20M and top 30 by Market Cap .

June Literature Research

Literature review on the **Kalman Filter** and **previous implementations** of the strategy.

Relevant Literature Areas:

1. Cointegration & Pairs Trading
2. Kalman Filter Hedge Ratio Estimation
3. Hong Kong/EM Volatility

July Analysis / Implementation

Implement pairs trading strategy using Python:

- Create a **reusable pipeline** for data processing, cointegration, pairs strategy, trading signal & backtesting.
- Key figures to collect: **hedge ratios** overtime, **trading signal** plots, **performance metrics**.

August Final Write Up

Aim to complete report by **early-mid August** to ensure deadline is reached.

Write up will highlight my **methodology, performance of a base strategy/ refined strategy, and limitations & extensions** to model developed.

PROJECT OVERVIEW - PROPOSED DATA UNIVERSE

Justification for sectors chosen:

- Selected tech and commodity-linked equities due to their relevance to Hong Kong and Mainland China.
- Chosen because they are exposed to different sources of volatility.
- Provides an initial equity universe for subsequent cointegration screening and pair formation.

Ticker (Tech)	Company (Tech)	Ticker (Energy/Materials)	Company (Energy/Materials)
700 HK Equity	TENCENT HOLDINGS LTD	857 HK Equity	PETROCHINA CO LTD-H
9988 HK Equity	ALIBABA GROUP HOLDING LTD	883 HK Equity	CNOOC LTD-H
1211 HK Equity	BYD CO LTD-H	1088 HK Equity	CHINA SHENHUA ENERGY CO-H
1810 HK Equity	XIAOMI CORP-CLASS B	2899 HK Equity	ZIJIN MINING GROUP CO LTD-H
300 HK Equity	MIDEA GROUP CO LTD-H	386 HK Equity	CHINA PETROLEUM & CHEMICAL-H
981 HK Equity	SEMICONDUCTOR MANUFACTURING-H	3993 HK Equity	CMOC GROUP LTD-H
9999 HK Equity	NETEASE INC	2259 HK Equity	ZIJIN GOLD INTERNATIONAL CO
3690 HK Equity	MEITUAN-CLASS B	1378 HK Equity	CHINA HONGQIAO GROUP LTD
9618 HK Equity	JD.COM INC-CLASS A	2600 HK Equity	ALUMINUM CORP OF CHINA LTD-H
9888 HK Equity	BAIDU INC-CLASS A	1898 HK Equity	CHINA COAL ENERGY CO-H
9961 HK Equity	TRIP.COM GROUP LTD	1772 HK Equity	GANFENG LITHIUM GROUP CO L-H
6690 HK Equity	HAIER SMART HOME CO LTD-H	1787 HK Equity	SHANDONG GOLD MINING CO LT-H
1024 HK Equity	KUAISHOU TECHNOLOGY	1171 HK Equity	YANKUANG ENERGY GROUP CO-H
1347 HK Equity	HUA HONG SEMICONDUCTOR LTD-H	358 HK Equity	JIANGXI COPPER CO LTD-H
2015 HK Equity	LI AUTO INC-CLASS A	1818 HK Equity	ZHAOJIN MINING INDUSTRY CO-H
6618 HK Equity	JD HEALTH INTERNATIONAL INC	1208 HK Equity	MMG LTD

Refer to Appendix for Full Data Universe

Thank you!

FURTHER QUESTIONS?

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APPENDIX - FULL INVESTMENT UNIVERSE

Ticker (Tech)	Company (Tech)	Ticker (Energy/Materials)	Company (Energy/Materials)
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992 HK Equity	LENOVO GROUP LTD	2099 HK Equity	CHINA GOLD INTERNATIONAL RES
9866 HK Equity	NIO INC-CLASS A	2883 HK Equity	CHINA OILFIELD SERVICES-H
9868 HK Equity	XPENG INC - CLASS A SHARES	2788 HK Equity	CHUANGXIN INDUSTRIES HOLDING
1698 HK Equity	TENCENT MUSIC ENT - CLASS A	3939 HK Equity	WANGUO GOLD GROUP LTD
9660 HK Equity	HORIZON ROBOTICS INC	3858 HK Equity	JIAXIN INTERNATIONAL RESOURC
20 HK Equity	SENSE TIME GROUP INC-CLASS B	1258 HK Equity	CHINA NONFERROUS MINING CORP
9626 HK Equity	BILIBILI INC-CLASS Z	3668 HK Equity	YANCOAL AUSTRALIA LTD
241 HK Equity	ALIBABA HEALTH INFORMATION TECH	3330 HK Equity	LINGBAO GOLD GROUP CO LTD-H
9863 HK Equity	ZHEJIANG LEAPMOTOR TECHNOLOGY-H	1164 HK Equity	CGN MINING CO LTD
2382 HK Equity	SUNNY OPTICAL TECH	2689 HK Equity	NINE DRAGONS PAPER HOLDINGS
285 HK Equity	BYD ELECTRONIC INTL CO LTD	2386 HK Equity	SINOPEC ENGINEERING GROUP-H
780 HK Equity	TONGCHENG TRAVEL HOLDINGS LTD	2610 HK Equity	NANSHAN ALUMINIUM INTERNATIO
3888 HK Equity	KINGSOFT CORP LTD	3899 HK Equity	CIMC ENRIC HOLDINGS LTD
268 HK Equity	KINGDEE INTERNATIONAL SOFTWARE	189 HK Equity	DONGYUE GROUP